Gullapalli Pranav anish

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.42
Sri Gayatri Junior College	Class XII (The Telangana State Board of Intermediate Education (TSBIE))	2021	93.60%
G Pulla Reddy High School	Class X (Board of Secondary Education, Telangana.)	2019	9.00%

SKILLS

Languages: Python, SQL

AI Domains: Computer Vision, NLP, Speech AI, RAG, LLM Integration, Multimodal AI, Generative AI

Backend & Systems: REST APIs, FastAPI, WebSockets, WebRTC, Microservices, Docker, Clean Architecture, SOLID Principles

Testing & Quality: Automated Unit & API Testing (Pytest)

Frameworks & Libraries: Scikit-learn, TensorFlow, PyTorch, OpenCV, YOLOv8, XGBoost, Pandas, NumPy, Plotly, TA-Lib

Databases: PostgreSQL, Redis, SQLite, SQLAlchemy

Data Science & ML Fundamentals: Predictive Modeling, Regression, Classification, Clustering, Model Evaluation, Hypothesis Testing

Tools: Git, GitHub, Flutter

PROJECTS

AI-Powered Smart Navigation Assistant for Educational Campuses -- M.Tech Thesis Project

- Collected real-world campus navigation and obstacle data through on-ground surveys across **IIT Hyderabad** to build a representative multimodal dataset for navigation and vision tasks.
- Developed a hybrid navigation framework combining **Google Routes API** with an **IIT-H campus graph** for **GPS**-based walking navigation and accessibility-aware campus routing.
- Built a real-time **YOLO26n** and **OpenCV** vision pipeline with **path-relevance and safety reasoning** to identify obstacles affecting the walking corridor while suppressing irrelevant detections.
- Integrated **Flutter**, **FastAPI**, **GPS**, **camera**, **TTS**, and **Multimodal AI** for voice-guided navigation and context-aware interpretation of selected visual scenes.
- Implemented **RAG-based campus knowledge retrieval**, **OCR-based sign recognition**, **depth estimation**, and **multimodal reasoning** for context-aware accessibility assistance and environmental understanding.

Tech Stack:*Python, FastAPI, Flutter, OpenCV, YOLOv8, OCR, SQLAlchemy, GPS, RAG, LLM Integration, Multimodal AI*

AI Phone Interpreter

- Built a **real-time multilingual voice interpreter** using Flutter and FastAPI, integrating **Azure Speech Services and Azure AI Translator** for two-way speech translation.
- Engineered an end-to-end **real-time speech pipeline** covering Speech-to-Text, language detection, neural machine translation, and Text-to-Speech, using **WebRTC for audio transport and WebSockets for signaling**.
- Designed a **scalable microservices architecture** with secure authentication, Redis-based messaging, and PostgreSQL-backed conversation persistence, supporting **Indian multilingual and code-mixed speech**.

Tech: *Flutter, FastAPI, Python, WebRTC, WebSockets, Azure Speech Services, Azure AI Translator, Docker, Redis, PostgreSQL*

AngelAI -- AI-Based Financial Signal Intelligence Platform

- Developed an end-to-end financial analytics platform using **Python**, **Angel One SmartAPI**, **Pandas**, **NumPy**, and **SQLite** to collect **NSE market data**, engineer technical features, generate trading signals, and track signal outcomes.
- Engineered 8 market features -- **RSI**, **EMA**, **MACD**, **Supertrend**, **Bollinger Bands**, **pivot levels**, volume, and price-action/trend-structure indicators -- to identify breakout and reversal opportunities.
- Developed and compared **Random Forest** and **XGBoost** classifiers on **188 historical signal outcomes**, achieving **0.84+** and **0.89+ cross-validation ROC-AUC**, respectively, for AI-based signal ranking.
- Applied **hypothesis testing**, **A/B testing**, **confidence intervals**, and **ANOVA** to evaluate performance differences between rule-based and AI-ranked signals.

Tech Stack: *Python, Streamlit, Plotly, SQLite, Angel One SmartAPI, Scikit-learn, XGBoost, Random Forest, Pandas, NumPy, TA-Lib, Telegram Bot API*

CERTIFICATIONS

- **Data Science & Machine Learning Certification** -- GeeksforGeeks

COURSE WORK

Machine Learning, Deep Learning, Linear Algebra & Calculus, Probability & Statistics, DBMS

EXTRACURRICULAR

- Competitive Table Tennis player -- Secured 2nd place at **Milan** (Inter-Hostel) and **Diesta** (Inter-Department) tournaments, IIT Hyderabad.
- Badminton enthusiast, playing regularly.